

Question			Marking details					Marks available					Prac																					
								AO1	AO2	AO3	Total	Maths																						
4	(a)	(i)	<table><tr><th>Particle</th><th>Symbol</th><th>Quark Combination</th><th>Charge/e</th><th>Baryon Number</th></tr><tr><td>Proton</td><td>p</td><td>uud</td><td>+1</td><td>1</td></tr><tr><td>Delta particle</td><td>Δ^{++}</td><td>uuu</td><td>+2</td><td>1</td></tr><tr><td>Electron</td><td>e^-</td><td>No quarks present</td><td>-1</td><td>0</td></tr><tr><td>Pion</td><td>π^-</td><td>$\bar{u}d$ or $d\bar{u}$</td><td>-1</td><td>0</td></tr></table> [3 × 1] marks for each correct row (ignoring row for proton)	Particle	Symbol	Quark Combination	Charge/e	Baryon Number	Proton	p	uud	+1	1	Delta particle	Δ^{++}	uuu	+2	1	Electron	e^-	No quarks present	-1	0	Pion	π^-	$\bar{u}d$ or $d\bar{u}$	-1	0		3		3		
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		(ii)	Electron	1			1																											
	(b)		Charge: LHS: -1 +1 (=0) RHS: -1 +2 -1 (=0) [must be shown how 0 is determined on both sides] (1) Lepton Number: LHS: 1 + 0 (= 1) RHS: 1 + 0 + 0 (=1) [must be shown how 1 is determined on both sides] (1)		2		2																											
	(c)	(i)	Up Quark: Δ^{++} contains 3 up quarks (uuu) = proton contains 2 up quarks (uud) + pion contains 1 up quark (ud) (1) Down Quark: Δ^{++} contains 0 down quarks = proton contains 1down quark (uud) + pion contains 1 antidown quark ($\bar{u}$). (1) Alternative for first mark: equation in quark form $uuu \rightarrow uud + \bar{u}d$		2		2																											
		(ii)	Any 2 × (1) from: - Very short lifetime / decays quickly / ref to 6×10^{-24} s - No change in u or d quark number (or flavour) - Only quarks or hadrons involved - No γ (or photon) or no neutrino involvement	2			2																											
	(d)		No practical use for electron or proton or both when they were discovered (1) Now many uses and one example given for either electron or proton (1)			2	2																											
			Question 4 total	3	7	2	12	0	0																									